

# NAZIRU HALILU

*AI · Computer Vision · Geospatial ML for Digital Agriculture*

Pamplona, Spain | +34 6 5296 7216 | [halilu.175366@e.unavarra.es](mailto:halilu.175366@e.unavarra.es)

[GitHub](#) | [ORCID](#) | [Zenodo records](#) | [Portfolio](#)

Agricultural engineer with a long-term research focus on the digitalization of agriculture, spanning computer vision, geospatial machine learning, agricultural control systems, and high-throughput phenotyping, grounded in a strong foundation in agricultural machinery, mechanization, and farm power systems. Current interests include self-supervised deep learning for crop-stress detection and vegetation-index estimation from UAV and multispectral imagery, and closed-loop sensing and control for irrigation, greenhouse automation, and precision farm machinery.

*This document is a complete, single-file capture of every section of the live portfolio site ([halilunaziru73-creator.github.io](https://halilunaziru73-creator.github.io)), including all gallery photographs and certificate images, generated for offline reading and archival purposes.*

## EDUCATION

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**Erasmus Mundus Joint Master's Degree (EMJMD), Cutting-Edge Technologies for Sustainable Agriculture** 2025 to expected 2027

*Public University of Navarre, Spain · UTAD, Portugal · Agricultural University of Athens, Greece*

**Bachelor of Engineering (B.Eng.), Agricultural and Bioresources Engineering, First Class Honours** 2019 to 2024

*Ahmadu Bello University, Zaria, Nigeria · CGPA 4.77/5.00, top 1% of graduating cohort*

**National Diploma, Agricultural Engineering Technology, Distinction** 2016 to 2018

*Ahmadu Bello University, Zaria, Nigeria · CGPA 3.78/4.00, top 1% of graduating cohort*

## RESEARCH EXPERIENCE

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**EMJMD Research Scholar (Mobility Track)** Sep 2026 to Jan 2027

*Agricultural University of Athens, Greece*

- Engineered NaCROP, a modular Python irrigation and crop-growth decision-support pipeline with a Tkinter GUI, validated across five smallholder crops around Zaria, Nigeria. [Code: GitHub](#)

**Summer Intern, Computer Vision Group** Jun 2026 to Aug 2026

*Dept. of Computer Engineering, Ahmadu Bello University, Zaria*

- Extended the GACL framework with four new geometry-invariant components, reaching 64.4% accuracy (macro AUC 0.882) across 19 disease classes. [Code: GitHub](#)

**EMJMD Research Scholar (Mobility Track)** Feb 2026 to Jun 2026

*University of Trás-os-Montes and Alto Douro (UTAD), Portugal*

- Integrated GIS, remote sensing, and ML pipelines across three cropping systems (R-squared 0.978–0.995), cutting lime demand by 2.69 tonnes. [Code: GitHub](#)
- Developed Multitask Singularity Regression Kriging (MSRK), reducing residual kriging uncertainty by roughly 20% across 17,864 soil samples. [Code: GitHub](#)
- Built an NDVI/logarithmic regression model forecasting grape ripening (R-squared = 0.9784) across a 6-hectare UTAD vineyard.
- Ran terrain characterisation and a 25-trap spatial monitoring network for grape moth across a 42.97-hectare Douro vineyard estate.

**EMJMD Research Scholar (Mobility Track)** Sep 2025 to Jan 2026

*Public University of Navarre, Spain*

- Built a 14-module open-source pipeline for RGB/multispectral vegetation indexing, GLCM texture analysis, and CNN/Faster R-CNN inference, reaching 33.8% balanced accuracy against a 4.55% chance baseline. [Code: GitHub](#)
- Designed STADOS, a five-stage radiometric harmonization framework, cutting temporal NDVI variance by 39% and 31% on Sentinel-2 imagery. [Code: GitHub](#)
- Built a Bayesian digital twin combining hydrological, hydraulic, and hybrid CNN-BiLSTM-attention sediment models for gully erosion biocontrol (hold-out R-squared = 0.90). [Code: GitHub](#)

**Undergraduate Project Assistantships** Mar 2024 to Mar 2025

*Depts. of Agricultural and Bioresources Engineering and Animal Science, ABU*

- Contributed data collection, statistical analysis, and literature review across concurrent undergraduate research projects spanning irrigation, soil erosion, postharvest, and animal systems.

**Project Assistant (Remote Data Lead), Smart Greenhouse Automation** 2024 to 2025

*IAR-ABU, Nigeria*

- Automated greenhouse environmental control by engineering adaptive irrigation-scheduling algorithms in Python.

**Student Research Assistant, Precision Crop Systems** 2022 to 2024

*NAERLS, Nigeria*

- Trained CNN-based crop-stress detection models in TensorFlow, reaching 78% classification accuracy on multispectral UAV imagery.

**Assistant Farm Manager (concurrent, part-time)** 2022 to 2024

*Yankari Nasarawa Farms, Nigeria*

- Managed automated climate-control and water-dosing systems across a commercial poultry facility.

### **Project Assistant, Smart Irrigation and Water Systems**

**2021 to 2022**

*BOMO-Basin Project, Nigeria*

- Automated water delivery across a one-hectare smart drip-irrigation array via closed-loop control logic.

### **Research Intern**

**2017 to 2020**

*NCAM, Ilorin and IAR, Zaria, Nigeria*

- Built Python data-logging pipelines benchmarking mechanization efficiency; supported baseline hydrological research.

### **Founder, Nhalilu's Biogas Plant and Sales Company**

*Registered with SMEDAN, Bioprocess Engineering*

- Founded and registered a biogas production and sales venture applying bioprocess engineering principles.

## PUBLICATIONS

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### Peer-Reviewed

- **Halilu, N.** (2026). A Hybrid CNN-BiLSTM-Attention Deep Learning Framework for Sediment Transport in an Agricultural Gully System. *IJRSI*, 13(7), 2920–2950. [DOI](#) | [Code](#)
- **Halilu, N., Sulaiman, J.** (2026). Real-Time RGB Proxy Vegetation Indexing and Texture Analysis for UAV and Handheld Crop Imagery. *IJRSI*, 13(7), 2951–2989. [DOI](#) | [Code](#)
- **Halilu, N., Sulaiman, J.** (2026). Sensing the Invisible: A Critical Review of NIR and Hyperspectral Analysis for Crop and Food Quality Assessment. *IJRSI*, 13(7), 2990–3020. [DOI](#)
- Ismail, K., **Halilu, N.** (2024). AI-Driven Precision Agriculture System for Optimizing Maize Yield in Northern Nigeria. *Proc. AfCPA*, 2(1), 100–120.
- Auwal, M., Yusuf, A., **Halilu, N.** (2020). Physio-chemical and bacteriological quality of water for irrigation purpose (Shika Dam). *Nigerian Research Journal of Engineering and Environmental Sciences*, 5(1), 198–204.

### Submitted / Under Review

- **Halilu, N., Sulaiman, J.** Geometry-Agnostic Contrastive Learning (GACL). *Biosystems Engineering* (Elsevier). Under review. [Code](#)
- **Halilu, N.** (2026). Digital Twin for the Evaluation of Experimental Gully Biocontrol Using Morning Glory. *Environmental Monitoring and Assessment*. Under review. [Code](#)
- **Halilu, N., Sulaiman, J.** Operationalizing GIS and Machine Learning across Contrasting Cropping Systems. *European Journal of Agronomy*. Under review. [Code](#)

### Preprints

- **Halilu, N.** (2026). Multitask Singularity Regression Kriging. *Zenodo Preprint*. [Code](#)
- **Halilu, N.** (2026). STADOS: Spectral-Temporal Adaptive Dark Object Subtraction. *Zenodo*. [Code](#)
- Halilu, N. (2025). A Systematic Review of IoT, Edge Computing, Time-Series Modelling and Control Systems in Precision Irrigation. *Zenodo*.

## RESEARCH PROJECTS AND OPEN-SOURCE REPOSITORIES

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- Geometry-Agnostic Contrastive Learning (GACL): <https://github.com/halilunaziru73-creator/Geometry-Agnostic-Contrastive-Learning-GACL>
- STADOS: Spectral-Temporal Adaptive Dark Object Subtraction: <https://github.com/halilunaziru73-creator/STADOS-Spectral-Temporal-Adaptive-Dark-Object-Subtraction>
- Multitask Singularity Regression Kriging (MSRK): <https://github.com/halilunaziru73-creator/Multitask-Singularity-Regression-Kriging-for-Spatio-Temporal-Soil-NPK-Crop-Stress-Yield-Prediction>
- Digital Twin for Gully Biocontrol Using Morning Glory: <https://github.com/halilunaziru73-creator/Digital-Twin-for-the-Evaluation-of-Experimental-Gully-Biocontrol-Using-Morning-Glory-Ipomoea-spp>
- Hybrid CNN-BiLSTM-Attention Sediment Transport Framework: <https://github.com/halilunaziru73-creator/Hybrid-CNN-BiLSTM-Attention-Sediment-Transport-Agricultural-Gully-System>
- Real-Time RGB Proxy Vegetation Indexing and Texture Analysis: <https://github.com/halilunaziru73-creator/Real-Time-RGB-Proxy-Vegetation-Indexing-and-Texture-Analysis-for-UAV-and-Handheld-Crop-Imagery>
- Operationalizing GIS and ML across Contrasting Cropping Systems: <https://github.com/halilunaziru73-creator/Operationalizing-GIS-and-Machine-Learning-across-Contrasting-Cropping-Systems>
- Geospatial Data Analysis (six-notebook GIS course): <https://github.com/halilunaziru73-creator/Geospatial-data-analysis>
- GIS-Based Livestock Slurry Suitability Delineation: <https://github.com/halilunaziru73-creator/GIS-based-delineation-of-areas-suitable-for-livestock-slurry-application>
- GIS Monitoring of Grape Moth (Lobesia botrana): <https://github.com/halilunaziru73-creator/GIS-Monitoring-of-Grape-Moth-Vineyard-Pest>

- GIS Monitoring of Vineyard Ripening (Brix): <https://github.com/halilunaziru73-creator/GIS-Monitoring-of-Vineyard-Ripening-Brix>
- NaCROP: Crop ET and Irrigation Decision-Support Pipeline: <https://github.com/halilunaziru73-creator/NaCROP>
- Integrated Decision-Support Model for Dairy Processing: <https://github.com/halilunaziru73-creator/Integrated-Decision-Support-Model-for-Dairy-Processing-under-Milk-and-Market-Variability>
- Full repository index: <https://github.com/halilunaziru73-creator>

## SKILLS

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**Geospatial AI / GIS:** QGIS, ArcGIS, ENVI, Google Earth Engine, Sentinel imagery, UAV imaging

**Machine Learning:** PyTorch, TensorFlow, Keras, scikit-learn, CNNs, self-supervised learning

**Crop Modelling:** APSIM Next Generation, crop growth simulation

**Climate AI:** CMIP6, ERA5, ACCESS-S2, downscaling and forecasting

**3D Computer Vision:** COLMAP, Open3D, Pix4D, Agisoft Metashape

**Programming:** Python, MATLAB, R, Git/GitHub, Zenodo

## TEACHING, LEADERSHIP AND COMMUNITY

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### Undergraduate Course Tutor

2020 to 2024

*ABU Zaria (NAMET)*

- Ran supplementary tutorials in Differential Equations and Transforms, Irrigation System Design, and Computer Applications in Agricultural Engineering.

### Secretary General, NAAESB

2023 to 2024

*National Assoc. of Agric. Engineering Students*

- Directed operational targets and inter-chapter scientific meetups for the national student body.

### Deputy Coordinator

2020 to 2025

*KOJYAF, Zango Shanu, Zaria*

- Coordinated tutorial and mentorship programmes preparing secondary students for JAMB entrance examinations.

### Director, Environmental Protection Agency

2023 to 2024

*ABU Students' Union Government*

- Directed campus environmental protection and sanitation initiatives.

### Physics and Mathematics Teacher

2022 to 2023

*Takai Academy, Zaria, Nigeria*

- Instructed senior classes in classical physics and computational principles.

## AWARDS AND HONOURS

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- Erasmus Mundus Joint Master Scholarship, European Union (2025)
- Best Project Award, ABU Senate (2024)
- Best Graduating Student, Agricultural and Bioresources Engineering (2024)
- Nigerian Federal Government Award for Academic Excellence (2024)
- Best Undergraduate Research Paper Award, ABU (2024)
- Best 300 Level Student, ABU (2022)
- Best 200 Level Student, ABU (2021)
- Award of Merit as Tutor, Faculty of Engineering, ABU (2021)
- Best Graduating Student, National Diploma, Samaru College of Agriculture (2019)
- PTDF Undergraduate Fellow (2023)
- Needs-Based University Fellowship (2022)
- Award of Honour, International Career Coaching Program, Envirofly Consulting under UK FCDO (2024)

- Award for Best Graduating Student, National Union for Engineering Student Association (2024)
- Award of Merit, Al-Azhar Institute of Arabic and Islamic Studies (2022)
- Award of Merit as Senator, Student Union Government, Samaru College of Agriculture (2017)
- Best Student in Science Subjects, Government Secondary School Jama'a (2014)

## CONFERENCES, SEMINARS AND WORKSHOPS

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- Data Management Workshop: From Phenotyping Experiment to FAIR Data, Norway (2026)
- Digital Agroecology Cluster Conference (EU-funded), Brussels (2025)
- 6th African Regional Conference on Irrigation and Drainage, ARCID (2025)
- Vibrational Spectroscopy and Chemometrics in Agro-Food Systems (EU-funded) (2025)
- Mechanistic Modeling of Rainfall-Driven Soil Solute Transport, University of Florida (2025)
- IEEE Intl. Conf. on Smart Agriculture Technologies (AgriTech), Valencia, Spain (2024)
- NAERLS Technical Seminar Series: AI in the Food Value Chain (2023)

## PROFESSIONAL MEMBERSHIPS

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American Society of Agricultural and Biological Engineers (ASABE) · Nigerian Institution of Agricultural Engineers (NIAE) · Nigerian Society of Engineers (NSE) · Pupil Engineer, COREN

## REFEREES

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**Prof. Javier Casalí Sarasibar**, Public University of Navarre, Spain · [jcs@unavarra.es](mailto:jcs@unavarra.es)

**Prof. Konstantinos Soulis**, Agricultural University of Athens, Greece · [soco@aua.gr](mailto:soco@aua.gr)

**Prof. Miguel Ángel Campo Bescós**, Public University of Navarre, Spain · [miguel.campo@unavarra.es](mailto:miguel.campo@unavarra.es)

**Prof. José Tadeu Marques Aranha**, UTAD, Portugal · [j\\_aranha@utad.pt](mailto:j_aranha@utad.pt)

**Dr. Ibrahim Babangida Dalha**, Ahmadu Bello University, Nigeria · [ibdalha@abu.edu.ng](mailto:ibdalha@abu.edu.ng)

**Dr. Habibu Ismail**, Ahmadu Bello University, Nigeria · [habfta@yahoo.com](mailto:habfta@yahoo.com)

**Dr. Mohammed Abdulsalam**, Ahmadu Bello University, Nigeria · [mabdulsalam@abu.edu.ng](mailto:mabdulsalam@abu.edu.ng)

**Dr. Yahya Gambo**, KFUPM, Saudi Arabia · [yahya.Gambo@kfupm.edu.sa](mailto:yahya.Gambo@kfupm.edu.sa)

# CERTIFICATIONS AND DOCUMENTED RECOGNITIONS

## Honorary Awards



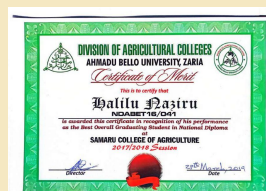
**Award of Merit, Senator (SRC/SUG)**

Samaru College of Agriculture  
SUG, 2017/2018 session



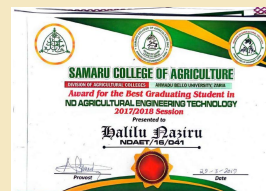
**Certificate of Merit, Best 200 Level Student**

Dept. of Agric. & Bio-Resources  
Engineering, ABU, 2018/2019  
session



**Best Overall Graduating Student, National Diploma**

Samaru College of Agriculture,  
ABU, 2017/2018 session



**Best Graduating Student, ND Agric. Engineering Technology**

Samaru College of Agriculture,  
ABU, 2017/2018 session

## Training



**Certificate of Continuing Professional Development**

IICCP, Envirofly Consulting (UK),  
60 CPD hours, 2024



**Tony Elumelu Foundation Certificate**

Business Management Training,  
2020



**Certificate of Attendance, Explore AI in Bioinformatics**

Genomac Institute Inc., USA, 2025



**Certificate of Learning, Genomics and Bioinformatics Bootcamp**

Genomac Institute Inc., USA, 2024



**International Webinar Series for Life Scientists**

Genomac Institute Inc., USA, 2024



**Bioinformatics Course 201**

Genomac Institute Inc., USA, 2024



**Genomics and Bioinformatics Short Course 301**

Genomac Institute Inc., USA, 2024



**SIWES Industrial Attachment Attestation**

NAERLS, ABU, 2023



**SIWES Industrial Attachment Attestation**

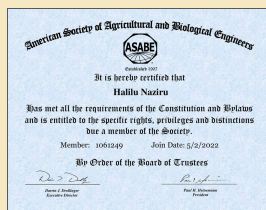
IAR, Samaru, ABU, 2017/2018



**Mechanistic Modeling Seminar Attendance**

ISFOOD Institute, UPNA, 2025

## Recognition



**ASABE Member Certificate**  
Member No. 1061249, joined 2022



**NIAE Student Member Certificate**  
Membership No. S2644, 2024



**Conference Paper Acceptance Letter**  
6th ARCID 2025, Abuja



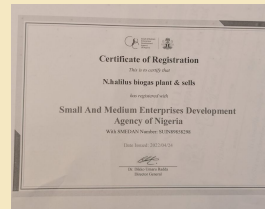
**IJRSI Publication Certificate**  
Sediment Transport paper, Vol. 13  
Issue 7, 2026



**IJRSI Publication Certificate**  
Real-Time RGB Vegetation  
Indexing paper, Vol. 13 Issue 7,  
2026



**IJRSI Publication Certificate**  
NIR/Hyperspectral review paper,  
Vol. 13 Issue 7, 2026



**SMEDAN Certificate of Registration**  
N.halilus Biogas Plant & Sells,  
2020



**Attestation to Proficiency in English Language**  
Ahmadu Bello University, 2024

## West African Examinations Council



**West African Senior School Certificate (WAEC)**  
Knowledge Is Power Secondary  
School, Zaria, 2015

## FULL ACADEMIC RECORD

Course-by-course record across National Diploma, B.Eng., and EMJMD programmes. Full authenticated transcripts are legalized by the Federal Ministry of Education, Nigeria, and the Consulado General de España en Lagos.

### ND Grading Scale

| Score %  | Letter Grade (LG) | Weighted Points (WP) | CGPA/Grade of Certificate |   |
|----------|-------------------|----------------------|---------------------------|---|
| 80 – 100 | A                 | 4                    | 3.5 – 4.0 Distinction     | A |
| 70 – 79  | AB                | 3.5                  | 3.0 – 3.49 Upper Credit   | B |
| 60 – 69  | B                 | 3                    | 2.5 – 2.99 Lower Credit   | C |
| 50 – 59  | BC                | 2.5                  | 2.0 – 2.49 Pass           | D |
| 40 – 49  | C                 | 2                    | Below 2.0 Fail            | F |
| 30 – 39  | CD                | 1.5                  |                           |   |
| 20 – 29  | D                 | 1                    |                           |   |
| 10 – 19  | E                 | 0.5                  |                           |   |
| 0 – 9    | F                 | 0                    |                           |   |

CGPA Achieved: 3.78/4.00, Distinction

### B.Eng Grading Scale

**Note**

1) **CLASSIFIED DEGREE**

| SCORE (%) | LETTER GRADE | GRADE POINT | CGPA        | CLASS OF DEGREE             |
|-----------|--------------|-------------|-------------|-----------------------------|
| 70 – 100  | A            | 5           | 4.50 - 5.00 | 1 <sup>st</sup> Class       |
| 60 – 69   | B            | 4           | 3.50 - 4.49 | 2 <sup>nd</sup> Class Upper |
| 50 – 49   | C            | 3           | 2.40 - 3.49 | 2 <sup>nd</sup> Class Lower |
| 45 – 49   | D            | 2           | 1.50 - 2.39 | Third Class                 |
| 40 – 44   | E            | 1           | 1.00 - 1.49 | Pass                        |
| 0-39      | F            | 0           | .00 - .99   | Fail                        |

2) **NON-CLASSIFIED DEGREE**  
(To be inserted by the Faculty concerned)

CONSULADO GENERAL DE ESPAÑA EN LAGOS  
Visto Bueno para legalizar la firma y/o el sello de **KINGSLEY ADIABAKPA**  
Ministerio de Asuntos Exteriores de Nigeria por ser, al

CGPA Achieved: 4.77/5.00, First Class

## Signatures and Official Seals

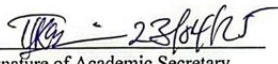
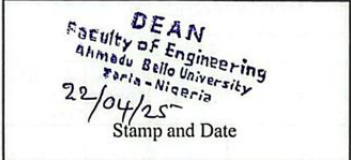
### Head of Department, Prof. Henry Igbadun

|         |                     |  |   |   |   |    |
|---------|---------------------|--|---|---|---|----|
| MMEN523 | Welding And Brazing |  | 2 | A | 5 | 10 |
|---------|---------------------|--|---|---|---|----|

Signature of Head of Department:   
 Stamp: 

Dept. of Agric & Bio-Resource Engineering, ABU, 2025/2026 session

### Dean, Engr. Prof. Jibril Muhammad Kaura & Academic Secretary

|                                                                                                                   |                                                                                                                         |                                  |          |
|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------|----------|
| 26/04/2025 25131010005259<br>65994,00 NGN CAYERO                                                                  | 101                                                                                                                     | 101                              | 7.11     |
|                                                                                                                   | Total Credit Units<br>(Registered)                                                                                      | ABU Cum Credit Units<br>(Earned) | ABU CGPA |
| <br>Signature of Dean          | <br>Signature of Academic Secretary |                                  |          |
| <br>22/04/25<br>Stamp and Date | <br>Stamp and Date                  |                                  |          |

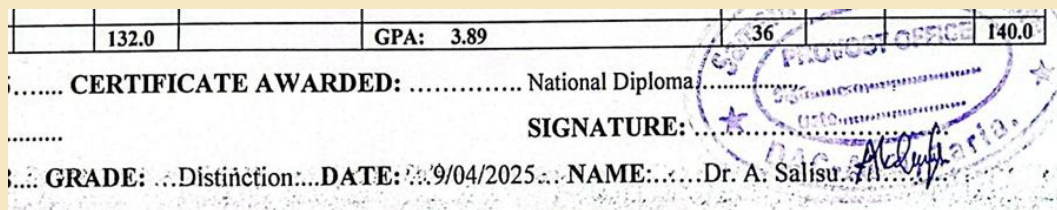
Faculty of Engineering, Ahmadu Bello University, dated 22/04/25

## Federal Ministry of Education & El Cónsul General, José María Ferré Trad



Federal Government of Nigeria & Consulado General de España en Lagos, dated 26 May 2025

### Dr. A. Salisu



Samaru College of Agriculture, ABU, National Diploma certificate, 2016 to 2018

### Course Categories (10 disciplines, full course-by-course marks on the live site)

- Agricultural Machinery, Mechanization & Power (10 courses)
- Soil, Water, Irrigation & Environmental Engineering (18 courses)
- Data Science, Precision Agriculture & Remote Sensing (12 courses)
- Processing, Post-Harvest & Biological Systems (8 courses)
- Mechanical & Materials Engineering (21 courses)
- Electrical, Instrumentation & Structures (10 courses)
- Mathematics, Statistics & Surveying (11 courses)
- Agronomy, Soil Science & Animal Sciences (4 courses)
- Management, Economics & General Studies (28 courses)
- Projects, Practical Exercises & Industrial Training (9 courses)

## PLACES WORKED AND STUDIED

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From Zaria, Nigeria, through Spain (Pamplona, Madrid, Barcelona), Portugal, Greece, Qatar, Morocco, Ethiopia, France, Italy, and Germany, over the course of research travel, conferences, and the Erasmus Mundus programme.



**10 Countries** across fieldwork, study, and the Erasmus Mundus programme.

## RECENT UPDATES

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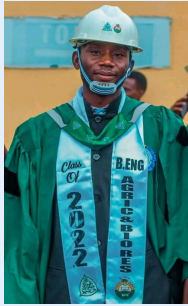
- Sep 2026: Published GIS monitoring datasets for grape moth and vineyard ripening (Brix), archived on Zenodo with DOIs.
- Aug 2026: Two papers accepted at IJRSI, covering sediment transport modelling and real-time vegetation indexing.
- Aug 2026: Archived STADOS on Zenodo, alongside updated releases of the Zaria irrigation DSS and NaCROP.
- Mar 2025: Paper accepted for presentation at the 6th ARCID, Abuja, on sediment transport rate prediction.

## GALLERY

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Field and campus photographs, organized by theme.

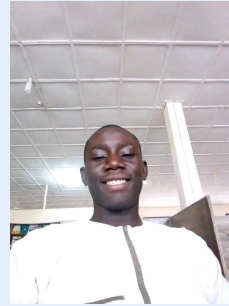
### Graduation Ceremony (3)



**B.Eng graduation ceremony, Class of 2022 cohort (graduated 2024 after COVID-19 and ASUU strike disruptions), Agric & Bio-Resources Engineering, 2019 to 2024**



**Graduation ceremony procession, Bachelor of Engineering, 2019 to 2024**



**National Diploma graduation celebration, Agricultural Engineering Technology, 2016 to 2018**

### EMJMD Programme & Welcome Events (10)



**Large group photo on a stone terrace, EMJMD cohort, 2025**



**Formal welcome event in a university corridor with staff and students, 2025**



**Group photo beside a metal sculpture installation, wind turbines in the distance, 2025**



**Receiving welcome bags on arrival at UTAD, 2026**



**Wearing an EU Global Gateway shirt for an EMJMD event, 2025**



**Outside the Agricultural University of Athens, 2026**



**At the Erasmus+ Pre-Departure Orientation event, 2025**



**In front of the UTAD banner, 2026**



**Beside the official SUSTAGRI  
EMJMD programme banner,  
2025**

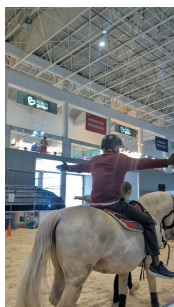


**Group photo with the EMJMD  
SUSTAGRI cohort, 2025**

## **Agricultural Fairs & Machinery Demonstrations (7)**



**Agricultural trade fair, rice and  
grain product stands, 2026**



**Horse-riding demonstration at  
an indoor arena, Forum Braga,  
2026**



**Dairy trade fair with the ALIP  
cow mascot, 2026**



**Farm machinery expo, compact  
tractor on display, 2026**



**Farm machinery expo, standing  
beside tractors, 2026**



**Trying a John Deere  
precision-agriculture GPS  
guidance system, 2026**



**Braga city tourism display at an  
expo, 2026**

## **Douro Valley Vineyard Fieldwork (7)**



**Group visit to Quinta do Seixo vineyard estate, Douro Valley, 2026**



**Visiting the Quinta S. Luiz port wine estate, Douro Valley, 2026**



**Irrigation sensor station on the vineyard terraces, 2026**



**Inside the wine fermentation tank hall, 2026**



**Walking between rows of stainless steel fermentation tanks, 2026**

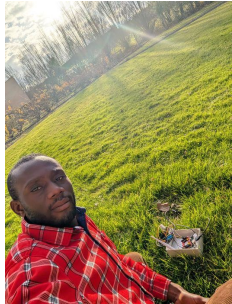


**Inspecting a weather station on the vineyard terraces, 2026**



**Fieldwork at the vineyard weather station, 2026**

### **Sensor & Field Research (3)**



**Testing a sensor prototype outdoors on the lawn, 2025**



**Inspecting a LoRa IoT irrigation sensor box in the field, 2025**



**Hyperspectral imaging fieldwork on apples, snow-covered orchard in winter, 2025**

### **Presentations & Conferences (3)**



**Classroom presentation on risks and limitations of spectral sensor deployments, 2025**



**Group selfie with conference lanyards, 2025**



**Presenting at a conference lectern, 2025**

## **Campus Life & Social Gatherings (11)**



**Amigo Secreto (Secret Santa) gift exchange in the classroom, 2025**



**Group celebration after the Secret Santa gift exchange, 2025**



**Evening gathering with classmates at an apartment party, 2025**



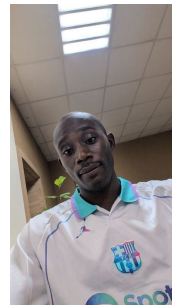
**Bowling and arcade outing with friends, 2025**



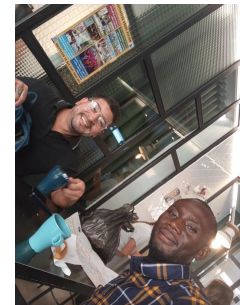
**Group walk near the Public University of Navarre, 2025**



**With friends, football jerseys, outdoor gathering, 2024**



**Casual indoor portrait, 2024**



**Coffee break with a friend on a balcony, 2026**



**Casual indoor portrait, 2025**



**Sharing a meal with a friend in the university cafeteria, 2025**



**Evening gathering with friends in the kitchen, 2025**

## **Portraits & Travel (10)**



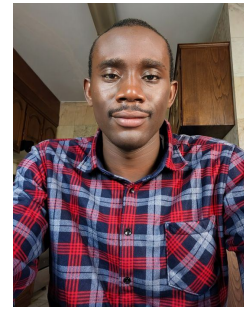
**Group hike, resting by a mountain stream, 2025**



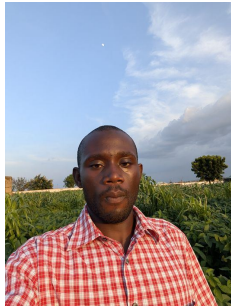
**Portrait on an Atlantic coast beach, 2026**



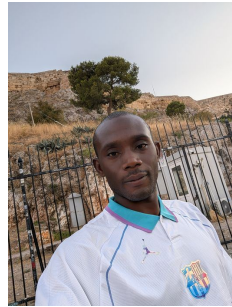
**Farm building with a solar panel canopy, 2024**



**Casual indoor portrait, 2024**



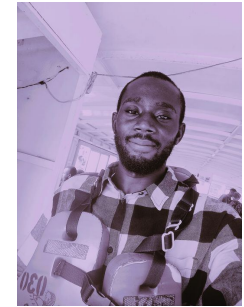
**Standing in a maize field at dusk, 2024**



**Near an ancient fortress wall in Greece, 2026**



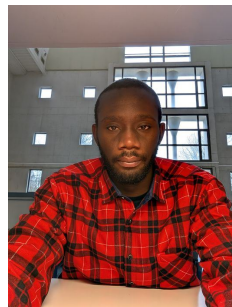
**Studying in the university library, 2026**



**Candid photo with a backpack, purple filter, 2025**



**Official student portrait for an event, 2025**



**Studying in a university building, 2026**

## **Greece — Cape Sounion (4)**



**Cape Sounion coastline, Greece, 2026**



**Temple of Poseidon ruins, Cape Sounion, Greece, 2026**



**Sunset over the Aegean Sea, Cape Sounion, Greece, 2026**



**Temple of Poseidon columns, Cape Sounion, Greece, 2026**

## CONTACT

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